Artificial Intelligence & Machine Learning Engineer (Mid-Level)

Opportunity

We are developing the next generation of predictive analytics and automated decisioning tools for our enterprise platform. As data volume and the demand for intelligent automation grow, we are expanding our core team to design, build, and deploy production-ready machine learning pipelines. This role matters because you will transform raw, unstructured data into actionable, intelligent features that directly optimize user workflows and drive measurable business outcomes.

This is a highly collaborative, hands-on engineering role. You will spend roughly 50% of your time designing and implementing data pipelines and training machine learning models, 30% of your time engineering cloud-native infrastructure to deploy these models into production, and 20% optimizing and maintaining existing ML services. The ideal candidate is an engineer at heart who pairs a strong understanding of statistical learning with the software engineering discipline required to deliver scalable, reliable AI solutions in the cloud.

Requirements

- 4+ years of professional software engineering experience, with at least 2+ years dedicated exclusively to building and deploying machine learning models in production
- Production-level mastery of Python and its core scientific computing stack (NumPy, Pandas, SciPy, Scikit-learn)
- Deep, hands-on experience with modern deep learning and machine learning frameworks, specifically PyTorch and TensorFlow
- 2+ years of experience with MLOps frameworks and workflow automation tools, such as MLflow, Kubeflow, or Weights & Biases
- Solid experience leveraging cloud-native AI/ML ecosystems, with a strong preference for AWS (Amazon SageMaker, AWS Bedrock, Lambda, and IAM) or equivalent services in GCP/Azure
- Experience designing and orchestrating automated data pipelines using Apache Airflow, Prefect, or AWS Step Functions
- Familiarity with fine-tuning and integrating Large Language Models (LLMs) using libraries like Hugging Face Transformers and orchestration frameworks like LangChain or LlamaIndex
- Proficiency with Docker containerization and deploying ML models as scalable APIs using frameworks like FastAPI or Flask
- Strong foundational understanding of SQL and relational databases, alongside Experience with NoSQL and Vector databases (such as pgvector, Pinecone, or Milvus)

Your Responsibilities

- Design, develop, evaluate, and deploy scalable machine learning, deep learning, and NLP models to solve complex predictive analysis and classification problems
- Architect and maintain robust MLOps and continuous training pipelines, ensuring seamless automated deployment, versioning, and tracking of model artifacts

- Build and optimize data ingestion and preprocessing workflows to handle both streaming and batch data inputs for training and inference
- Containerize machine learning models and deploy them as low-latency, high-availability microservices or serverless functions in the cloud
- Implement model monitoring, telemetry, and logging infrastructures to track data drift, concept drift, and overall model performance in real time
- Partner with data scientists, software engineers, and product stakeholders to translate strategic business requirements into scalable, reliable technical solutions
- Write comprehensive unit and integration tests for data schemas, pipeline logic, and model execution steps to maintain high code quality

You'll Love

- Taking models out of theoretical research environments and scaling them into production systems that impact thousands of daily users
- Working with a cutting-edge tech stack built around modern cloud infrastructure, advanced MLOps tools, and best-in-class deep learning frameworks
- Collaborating with a fast-paced, highly collaborative team that deeply values scientific experimentation, architectural rigor, and clean code
- Continually expanding your skillset by exploring and implementing the latest advancements in generative AI and foundational model patterns

How You Work

- You balance research and engineering—you approach model selection with scientific curiosity but stay relentlessly focused on production viability and system performance
- You are self-motivated and thrive in environments with minimal process overhead, relying on automation and collaborative documentation to maintain alignment
- You are a clear and concise communicator who can effortlessly explain abstract algorithmic trade-offs or complex data architectures to non-technical teammates
- You value engineering excellence, treating model code, configurations, and data definitions with the same architectural discipline as core application infrastructure

What Success Looks Like

- Within the first 90 days, you have successfully automated an existing manual model training process using SageMaker and MLflow, reducing deployment cycle times by 50%
- The real-time inference microservices you build consistently maintain latency profiles under 100ms under simulated production peak traffic conditions
- You successfully stand up an automated monitoring pipeline that detects feature drift and alerts the team before model performance degrades in production
- The team adopts your data processing design patterns, significantly accelerating the feature engineering lifecycle for subsequent machine learning initiatives